

Interpretable Machine Learning

Feature Importance

Permutation Feature Importance (PFI)

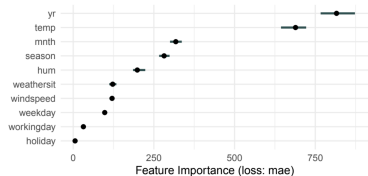
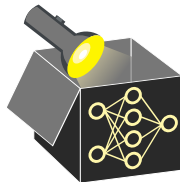


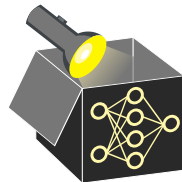
Figure: Bike Sharing Dataset

Learning goals

- Understand how PFI is computed
- Understanding strengths and weaknesses

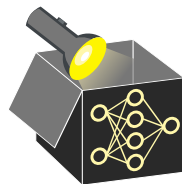
MOTIVATION FOR PFI

- **Goal:** Assess how important feature(s) X_S are for predictive performance of a **fixed trained model** $\hat{f}$ on a given dataset $\mathcal{D}$
- **Idea:** Estimate performance change when X_S is "made uninformative"

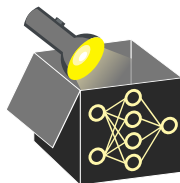


MOTIVATION FOR PFI

- **Goal:** Assess how important feature(s) X_S are for predictive performance of a **fixed trained model** $\hat{f}$ on a given dataset $\mathcal{D}$
- **Idea:** Estimate performance change when X_S is "made uninformative"
- **Question:** Can we make X_S uninformative by removing it from model?
 $\rightsquigarrow$ No, $\hat{f}$ was trained with X_S ; retraining without X_S gives a different model



MOTIVATION FOR PFI



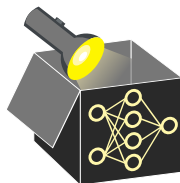
- **Goal:** Assess how important feature(s) X_S are for predictive performance of a **fixed trained model** $\hat{f}$ on a given dataset $\mathcal{D}$
- **Idea:** Estimate performance change when X_S is "made uninformative"
- **Question:** Can we make X_S uninformative by removing it from model?
 $\rightsquigarrow$ No, $\hat{f}$ was trained with X_S ; retraining without X_S gives a different model
- **Solution:** Simulate feature removal by replacing X_S with a perturbed version $\tilde{X}_S$ that is independent of (X_{-S}, Y) but preserves distrib. $\mathbb{P}(X_S)$
 $\rightsquigarrow$ Compare **baseline predictions** $\hat{f}(X)$ with **perturbed predictions** $\hat{f}(\tilde{X}_S, X_{-S})$

$$\text{PFI}_S := \underbrace{\mathbb{E}\left[L(\hat{f}(\tilde{X}_S, X_{-S}), Y)\right]}_{\text{risk after "destroying" } X_S} - \underbrace{\mathbb{E}\left[L(\hat{f}(X), Y)\right]}_{\text{baseline risk}},$$

- **How to perturb X_S ?**
 - Add random noise: distorts $\mathbb{P}(X_S)$ (not used)
 - Permutation: preserves marginal $\mathbb{P}(X_S)$, breaks dependence with Y (used)

PERMUTATION FEATURE IMPORTANCE (PFI)

► "Breiman" 2001



Sample estimator (using independent test set $\mathcal{D} = \{(\mathbf{x}^{(i)}, y^{(i)})\}_{i=1}^n$)

- Measure error **with feat. values** $\mathbf{x}_S$ and **with permuted feat. values** $\tilde{\mathbf{x}}_S$
- Repeat permutation (e.g., m times) and average difference of both errors:

$$\widehat{PFI}_S = \frac{1}{m} \sum_{k=1}^m [\mathcal{R}_{\text{emp}}(\hat{f}, \tilde{\mathcal{D}}_{(k)}^S) - \mathcal{R}_{\text{emp}}(\hat{f}, \mathcal{D})]$$

- $\mathcal{D}_S^{(k)}$: dataset with column(s) $\mathbf{x}_S$ are **permuted** once (in repetition k)
- $\mathcal{R}_{\text{emp}}(\hat{f}, \mathcal{D}) = \frac{1}{n} \sum_{(x,y) \in \mathcal{D}} L(\hat{f}(x), y)$: Measures performance of $\hat{f}$ using $\mathcal{D}$
- Average over m permutations to reduce Monte-Carlo variance

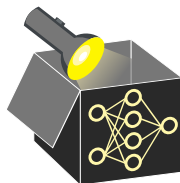
Example of permuting feature $\mathbf{x}_S$ with $S = \{1\}$ and $m = 6$ permutations:

$\mathcal{D}$	$\Rightarrow$	$\tilde{\mathcal{D}}_{(1)}^S$	$\tilde{\mathcal{D}}_{(2)}^S$	$\tilde{\mathcal{D}}_{(3)}^S$	$\tilde{\mathcal{D}}_{(4)}^S$	$\tilde{\mathcal{D}}_{(5)}^S$	$\tilde{\mathcal{D}}_{(6)}^S$																																																																																				
<table><tr><th>$\mathbf{x}_1$</th><th>$\mathbf{x}_2$</th><th>$\mathbf{x}_3$</th></tr><tr><td>1</td><td>4</td><td>7</td></tr><tr><td>2</td><td>5</td><td>8</td></tr><tr><td>3</td><td>6</td><td>9</td></tr></table>	$\mathbf{x}_1$	$\mathbf{x}_2$	$\mathbf{x}_3$	1	4	7	2	5	8	3	6	9		<table><tr><th>$\mathbf{x}_S$</th><th>$\mathbf{x}_2$</th><th>$\mathbf{x}_3$</th></tr><tr><td>1</td><td>4</td><td>7</td></tr><tr><td>2</td><td>5</td><td>8</td></tr><tr><td>3</td><td>6</td><td>9</td></tr></table>	$\mathbf{x}_S$	$\mathbf{x}_2$	$\mathbf{x}_3$	1	4	7	2	5	8	3	6	9	<table><tr><th>$\mathbf{x}_S$</th><th>$\mathbf{x}_2$</th><th>$\mathbf{x}_3$</th></tr><tr><td>2</td><td>4</td><td>7</td></tr><tr><td>1</td><td>5</td><td>8</td></tr><tr><td>3</td><td>6</td><td>9</td></tr></table>	$\mathbf{x}_S$	$\mathbf{x}_2$	$\mathbf{x}_3$	2	4	7	1	5	8	3	6	9	<table><tr><th>$\mathbf{x}_S$</th><th>$\mathbf{x}_2$</th><th>$\mathbf{x}_3$</th></tr><tr><td>2</td><td>4</td><td>7</td></tr><tr><td>3</td><td>5</td><td>8</td></tr><tr><td>1</td><td>6</td><td>9</td></tr></table>	$\mathbf{x}_S$	$\mathbf{x}_2$	$\mathbf{x}_3$	2	4	7	3	5	8	1	6	9	<table><tr><th>$\mathbf{x}_S$</th><th>$\mathbf{x}_2$</th><th>$\mathbf{x}_3$</th></tr><tr><td>1</td><td>4</td><td>7</td></tr><tr><td>3</td><td>5</td><td>8</td></tr><tr><td>2</td><td>6</td><td>9</td></tr></table>	$\mathbf{x}_S$	$\mathbf{x}_2$	$\mathbf{x}_3$	1	4	7	3	5	8	2	6	9	<table><tr><th>$\mathbf{x}_S$</th><th>$\mathbf{x}_2$</th><th>$\mathbf{x}_3$</th></tr><tr><td>3</td><td>4</td><td>7</td></tr><tr><td>1</td><td>5</td><td>8</td></tr><tr><td>2</td><td>6</td><td>9</td></tr></table>	$\mathbf{x}_S$	$\mathbf{x}_2$	$\mathbf{x}_3$	3	4	7	1	5	8	2	6	9	<table><tr><th>$\mathbf{x}_S$</th><th>$\mathbf{x}_2$</th><th>$\mathbf{x}_3$</th></tr><tr><td>3</td><td>4</td><td>7</td></tr><tr><td>2</td><td>5</td><td>8</td></tr><tr><td>1</td><td>6</td><td>9</td></tr></table>	$\mathbf{x}_S$	$\mathbf{x}_2$	$\mathbf{x}_3$	3	4	7	2	5	8	1	6	9
$\mathbf{x}_1$	$\mathbf{x}_2$	$\mathbf{x}_3$																																																																																									
1	4	7																																																																																									
2	5	8																																																																																									
3	6	9																																																																																									
$\mathbf{x}_S$	$\mathbf{x}_2$	$\mathbf{x}_3$																																																																																									
1	4	7																																																																																									
2	5	8																																																																																									
3	6	9																																																																																									
$\mathbf{x}_S$	$\mathbf{x}_2$	$\mathbf{x}_3$																																																																																									
2	4	7																																																																																									
1	5	8																																																																																									
3	6	9																																																																																									
$\mathbf{x}_S$	$\mathbf{x}_2$	$\mathbf{x}_3$																																																																																									
2	4	7																																																																																									
3	5	8																																																																																									
1	6	9																																																																																									
$\mathbf{x}_S$	$\mathbf{x}_2$	$\mathbf{x}_3$																																																																																									
1	4	7																																																																																									
3	5	8																																																																																									
2	6	9																																																																																									
$\mathbf{x}_S$	$\mathbf{x}_2$	$\mathbf{x}_3$																																																																																									
3	4	7																																																																																									
1	5	8																																																																																									
2	6	9																																																																																									
$\mathbf{x}_S$	$\mathbf{x}_2$	$\mathbf{x}_3$																																																																																									
3	4	7																																																																																									
2	5	8																																																																																									
1	6	9																																																																																									

Note: S refers to a subset of features, here $|S| = 1$ to measure impact of permuting $\mathbf{x}_1$ on performance

PERMUTATION FEATURE IMPORTANCE

	$\tilde{\mathcal{D}}_{(k)}^S$			$\mathcal{D}$		
i	$\mathbf{x}_S$	$\mathbf{x}_2$	$\mathbf{x}_3$	$\mathbf{x}_1$	$\mathbf{x}_2$	$\mathbf{x}_3$
1	2	4	7	1	4	7
$\vdots$	1	5	8	2	5	8
n	3	6	9	3	6	9



1. Perturbation: Sample feature values from the distribution of x_S ($P(X_S)$).

⇒ Randomly permute feature x_S

⇒ Replace x_S with permuted feat. $\tilde{x}_S$ and create data $\tilde{\mathcal{D}}^S$ containing $\tilde{x}_S$

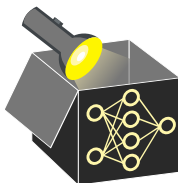
PERMUTATION FEATURE IMPORTANCE

	$\tilde{\mathcal{D}}_{(k)}^S$			$\mathcal{D}$		
i	$\mathbf{x}_S$	$\mathbf{x}_2$	$\mathbf{x}_3$	$\mathbf{x}_1$	$\mathbf{x}_2$	$\mathbf{x}_3$
1	2	4	7	1	4	7
$\vdots$	1	5	8	2	5	8
n	3	6	9	3	6	9

$\hat{f}$
0.6
0.6
0.6

$\hat{f}$
0.4
0.8
0.6

- 1. Perturbation:** Sample feature values from the distribution of x_S ($P(X_S)$).
 - $\Rightarrow$ Randomly permute feature x_S
 - $\Rightarrow$ Replace x_S with permuted feat. $\tilde{x}_S$ and create data $\tilde{\mathcal{D}}^S$ containing $\tilde{x}_S$
- 2. Prediction:** Make predictions for both data, i.e., $\mathcal{D}$ and $\tilde{\mathcal{D}}^S$

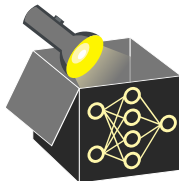


PERMUTATION FEATURE IMPORTANCE

	$\tilde{\mathcal{D}}_{(k)}^S$			$\mathcal{D}$		
i	$\mathbf{x}_1$	$\mathbf{x}_2$	$\mathbf{x}_3$	$\mathbf{x}_1$	$\mathbf{x}_2$	$\mathbf{x}_3$
1	2	4	7	1	4	7
$\vdots$	1	5	8	2	5	8
n	3	6	9	3	6	9

$L(\hat{f}, y)$	
	0.9
	0.5
	0.1

$L(\hat{f}, y)$	
	0.25
	0.35
	0.1



3. Aggregation:

- Compute the loss for each observation in both data sets

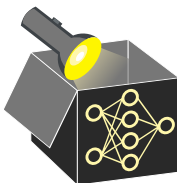
PERMUTATION FEATURE IMPORTANCE

	$\tilde{\mathcal{D}}_{(k)}^S$			$\mathcal{D}$			
i	$\mathbf{x}_s$	$\mathbf{x}_2$	$\mathbf{x}_3$	$\mathbf{x}_1$	$\mathbf{x}_2$	$\mathbf{x}_3$	ΔL
1	2	4	7	1	4	7	0.65
$\vdots$	1	5	8	2	5	8	0.15
n	3	6	9	3	6	9	0

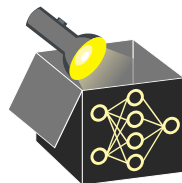
$L(\hat{f}, y)$		$L(\hat{f}, y)$
0.9	-	0.25
0.5		0.35
0.1		0.1

3. Aggregation:

- Compute the loss for each observation in both data sets
- Take the difference of both losses ΔL for each observation



PERMUTATION FEATURE IMPORTANCE



$$\mathcal{R}_{\text{emp}}(\hat{f}, \tilde{\mathcal{D}}_{(k)}^S) - \mathcal{R}_{\text{emp}}(\hat{f}, \mathcal{D})$$

i	$\mathbf{x}_S$	$\mathbf{x}_2$	$\mathbf{x}_3$	$\mathbf{x}_1$	$\mathbf{x}_2$	$\mathbf{x}_3$	ΔL
1	2	4	7	1	4	7	0.65
$\vdots$	1	5	8	2	5	8	0.15
n	3	6	9	3	6	9	0

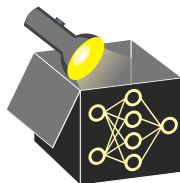
$= 0.267$

3. Aggregation:

- Compute the loss for each observation in both data sets
- Take the difference of both losses ΔL for each observation
- Average this change in loss across all observations

Note: Same as computing $\mathcal{R}_{\text{emp}}$ on both data sets and taking difference

PERMUTATION FEATURE IMPORTANCE



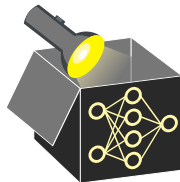
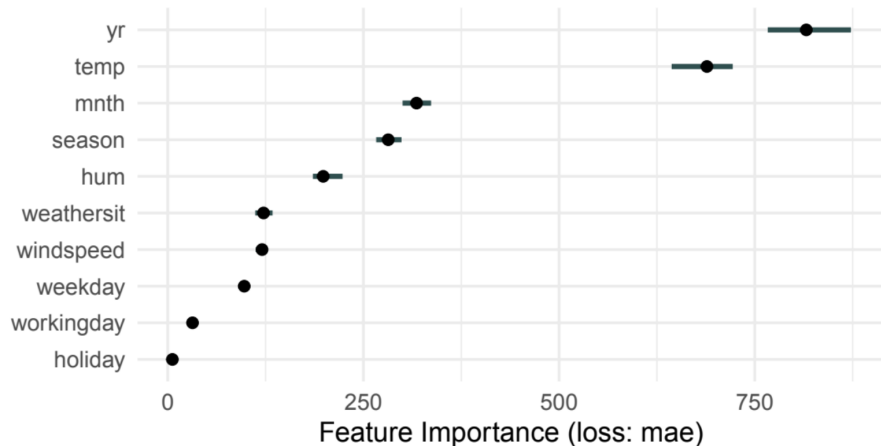
$$\mathcal{R}_{\text{emp}}(\hat{f}, \tilde{\mathcal{D}}_{(k)}^S) - \mathcal{R}_{\text{emp}}(\hat{f}, \mathcal{D})$$

1	i	$\mathbf{x}_S$	$\mathbf{x}_2$	$\mathbf{x}_3$	$\mathbf{x}_1$	$\mathbf{x}_2$	$\mathbf{x}_3$	ΔL	$= 0.267$ $\widehat{PFI}_S = \frac{1}{2} (0.267 + 0.4)$
	1	2	4	7	1	4	7	0.65	
	$\vdots$	1	5	8	2	5	8	0.15	
	n	3	6	9	3	6	9	0	
m	i	$\mathbf{x}_S$	$\mathbf{x}_2$	$\mathbf{x}_3$	$\mathbf{x}_1$	$\mathbf{x}_2$	$\mathbf{x}_3$	ΔL	
	1	3	4	7	1	4	7	0.85	$= 0.4$
	$\vdots$	2	5	8	2	5	8	0	
	n	1	6	9	3	6	9	0.35	

3. Aggregation:

- Compute the loss for each observation in both data sets
- Take the difference of both losses ΔL for each observation
- Average this change in loss across all observations
- Repeat perturbation and average over multiple repetitions

EXAMPLE: BIKE SHARING DATASET

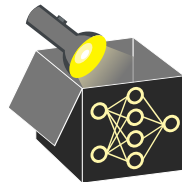


Interpretation:

- yr and temp are most important feats using mean absolute error (MAE)
- Destroying info. about yr by permuting it increases MAE of model by 816
- Error bars show 5% and 95% quantiles over multiple permutations

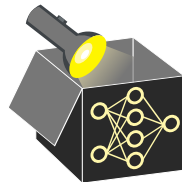
COMMENTS ON PFI

- Interpretation: Increase in error when feature's information is destroyed



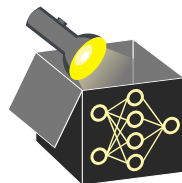
COMMENTS ON PFI

- Interpretation: Increase in error when feature's information is destroyed
- Results can be unreliable due to random permutations
⇒ Solution: Average results over multiple repetitions



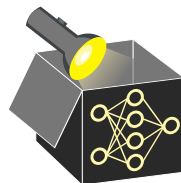
COMMENTS ON PFI

- Interpretation: Increase in error when feature's information is destroyed
- Results can be unreliable due to random permutations
⇒ Solution: Average results over multiple repetitions
- Permuting features despite correlation/dependence with other features can lead to unrealistic combinations of feature values
⇒ Extrapolation issue



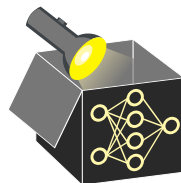
COMMENTS ON PFI

- Interpretation: Increase in error when feature's information is destroyed
- Results can be unreliable due to random permutations
⇒ Solution: Average results over multiple repetitions
- Permuting features despite correlation/dependence with other features can lead to unrealistic combinations of feature values
⇒ Extrapolation issue
- PFI automatically includes importance of interaction effects with other features
⇒ Permuting x_j also destroys interactions with permuted feature
⇒ PFI score contains importance of all interactions with permuted feature



COMMENTS ON PFI

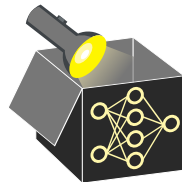
- Interpretation: Increase in error when feature's information is destroyed
- Results can be unreliable due to random permutations
⇒ Solution: Average results over multiple repetitions
- Permuting features despite correlation/dependence with other features can lead to unrealistic combinations of feature values
⇒ Extrapolation issue
- PFI automatically includes importance of interaction effects with other features
⇒ Permuting x_j also destroys interactions with permuted feature
⇒ PFI score contains importance of all interactions with permuted feature
- Interpretation of PFI depends on whether training or test data is used



COMMENTS ON PFI - EXTRAPOLATION

Example: Let $y = x_3 + \epsilon_y$, with $\epsilon_y \sim \mathcal{N}(0, 0.1)$.

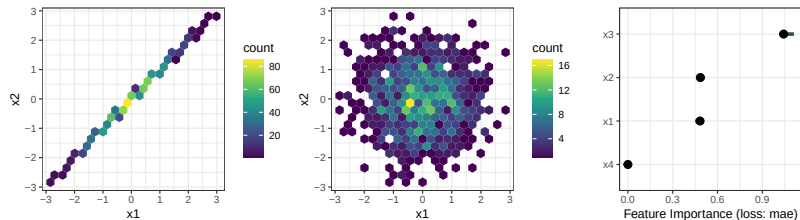
- $x_1 := \epsilon_1$, $x_2 := x_1 + \epsilon_2$; highly correlated ($\epsilon_1 \sim \mathcal{N}(0, 1)$, $\epsilon_2 \sim \mathcal{N}(0, 0.01)$)
- $x_3 := \epsilon_3$, $x_4 := \epsilon_4$, with $\epsilon_3, \epsilon_4 \sim \mathcal{N}(0, 1)$; all noise terms ϵ_j are indep.
- Fitting a linear model yields $\hat{f}(\mathbf{x}) \approx 0.3x_1 - 0.3x_2 + x_3$



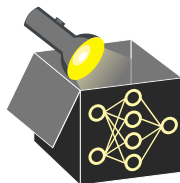
COMMENTS ON PFI - EXTRAPOLATION

Example: Let $y = x_3 + \epsilon_y$, with $\epsilon_y \sim \mathcal{N}(0, 0.1)$.

- $x_1 := \epsilon_1$, $x_2 := x_1 + \epsilon_2$; highly correlated ($\epsilon_1 \sim \mathcal{N}(0, 1)$, $\epsilon_2 \sim \mathcal{N}(0, 0.01)$)
- $x_3 := \epsilon_3$, $x_4 := \epsilon_4$, with $\epsilon_3, \epsilon_4 \sim \mathcal{N}(0, 1)$; all noise terms ϵ_j are indep.
- Fitting a linear model yields $\hat{f}(\mathbf{x}) \approx 0.3x_1 - 0.3x_2 + x_3$



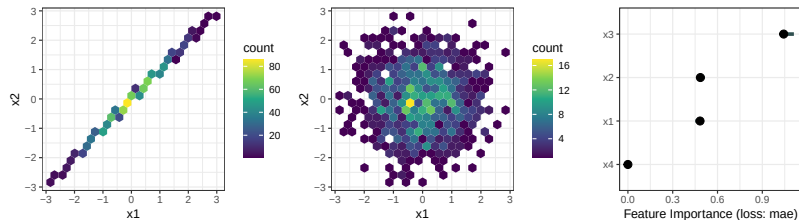
Hexbin plot of (x_1, x_2) before (left) and after (center) permuting x_1 ;
PFI scores (right).



COMMENTS ON PFI - EXTRAPOLATION

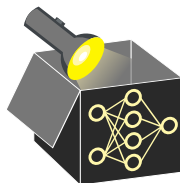
Example: Let $y = x_3 + \epsilon_y$, with $\epsilon_y \sim \mathcal{N}(0, 0.1)$.

- $x_1 := \epsilon_1$, $x_2 := x_1 + \epsilon_2$; highly correlated ($\epsilon_1 \sim \mathcal{N}(0, 1)$, $\epsilon_2 \sim \mathcal{N}(0, 0.01)$)
- $x_3 := \epsilon_3$, $x_4 := \epsilon_4$, with $\epsilon_3, \epsilon_4 \sim \mathcal{N}(0, 1)$; all noise terms ϵ_j are indep.
- Fitting a linear model yields $\hat{f}(\mathbf{x}) \approx 0.3x_1 - 0.3x_2 + x_3$



Hexbin plot of (x_1, x_2) before (left) and after (center) permuting x_1 ;
PFI scores (right).

- $\Rightarrow x_1, x_2$ cancel in $\hat{f}$ since $x_1 \approx x_2$, hence $0.3x_1 - 0.3x_2 \approx 0$
 $\rightsquigarrow$ should be irrelevant
- $\Rightarrow$ Permuting x_1 breaks joint structure $\rightsquigarrow$ unrealistic inputs
- $\Rightarrow PFI > 0$ due to extrapolation (PFI evaluates model on unrealistic inputs)
 $\rightsquigarrow x_1, x_2$ are misleadingly considered relevant

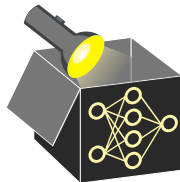


COMMENTS ON PFI - INTERACTIONS

Example: Let $x_1, \dots, x_4$ be independently and uniformly sampled from $\{-1, 1\}$ and

$$y := x_1 x_2 + x_3 + \epsilon_Y \text{ with } \epsilon_Y \sim N(0, 1)$$

Fitting a LM yields $\hat{f}(x) \approx x_1 x_2 + x_3$.



COMMENTS ON PFI - INTERACTIONS

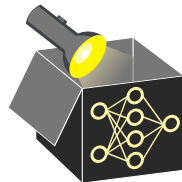
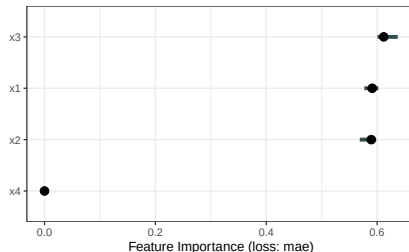
Example: Let $x_1, \dots, x_4$ be independently and uniformly sampled from $\{-1, 1\}$ and

$$y := x_1 x_2 + x_3 + \epsilon_Y \text{ with } \epsilon_Y \sim N(0, 1)$$

Fitting a LM yields $\hat{f}(x) \approx x_1 x_2 + x_3$.

Although x_3 alone contributes as much to the prediction as x_1 and x_2 jointly, all three are considered equally relevant.

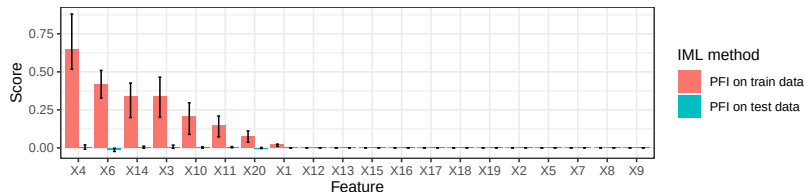
⇒ PFI does not fairly attribute the performance to the individual features.



COMMENTS ON PFI - TRAIN VS. TEST DATA

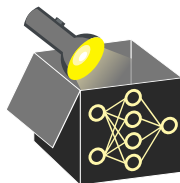
Example:

- $x_1, \dots, x_{20}, y$ are independently sampled from $\mathcal{U}(-10, 10)$
- Train set: $n = 50$ (intentionally small) and large test set
- Model: `xgboost` with default settings (overfits strongly)



Observation:

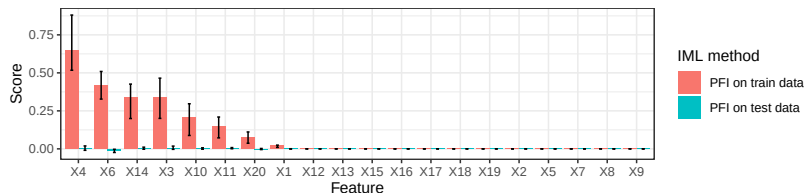
- PFI on train data highlights features that the model overfitted to.
- PFI on test data detects no relevant features.



COMMENTS ON PFI - TRAIN VS. TEST DATA

Example:

- $x_1, \dots, x_{20}, y$ are independently sampled from $\mathcal{U}(-10, 10)$
- Train set: $n = 50$ (intentionally small) and large test set
- Model: `xgboost` with default settings (overfits strongly)

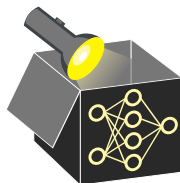


Observation:

- PFI on train data highlights features that the model overfitted to.
- PFI on test data detects no relevant features.

Why? $PFI \neq 0$ if permuting a feature breaks a dependency the model relies on. Model overfits due to spurious feature-target dependencies in train that vanish on test.

⇒ To find features that help the model to generalize, compute PFI on test data.

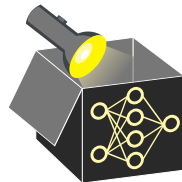


IMPLICATIONS OF PFI

Can we get insight into whether the ...

❶ feature x_j is causal for the prediction?

- $PFI_j \neq 0 \Rightarrow$ model relies on x_j
- As the train vs. test data example shows, the converse does not hold



IMPLICATIONS OF PFI

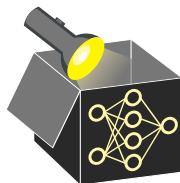
Can we get insight into whether the ...

❶ feature x_j is causal for the prediction?

- $PFI_j \neq 0 \Rightarrow$ model relies on x_j
- As the train vs. test data example shows, the converse does not hold

❷ feature x_j contains prediction-relevant information?

- $PFI_j \neq 0 \Rightarrow x_j$ is dependent on y , x_{-j} , or both (due to extrapolation)
- x_j is not exploited by model (regardless of its usefulness for y)
 $\Rightarrow PFI_j = 0$



IMPLICATIONS OF PFI

Can we get insight into whether the ...

- ❶ feature x_j is causal for the prediction?
 - $PFI_j \neq 0 \Rightarrow$ model relies on x_j
 - As the train vs. test data example shows, the converse does not hold
- ❷ feature x_j contains prediction-relevant information?
 - $PFI_j \neq 0 \Rightarrow x_j$ is dependent on y , x_{-j} , or both (due to extrapolation)
 - x_j is not exploited by model (regardless of its usefulness for y)
 $\Rightarrow PFI_j = 0$
- ❸ model requires access to x_j to achieve its prediction performance?
 - As shown by the extrapolation example, such insight is not possible

